

Geometry-based Methods in Vision

CMU 16-822, Fall 2024

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Administrivia

- Assignment 0 out
 - *Due on Thu next week*
 - *Please let us know if you face any issues in uploading webpages*
- Problem Set 0 out
 - *Also due on Thu next week*
 - *40pts max (of 48 total) — aimed at getting you to brush up linear algebra*

Lecture 2: (2D) Projective Geometry

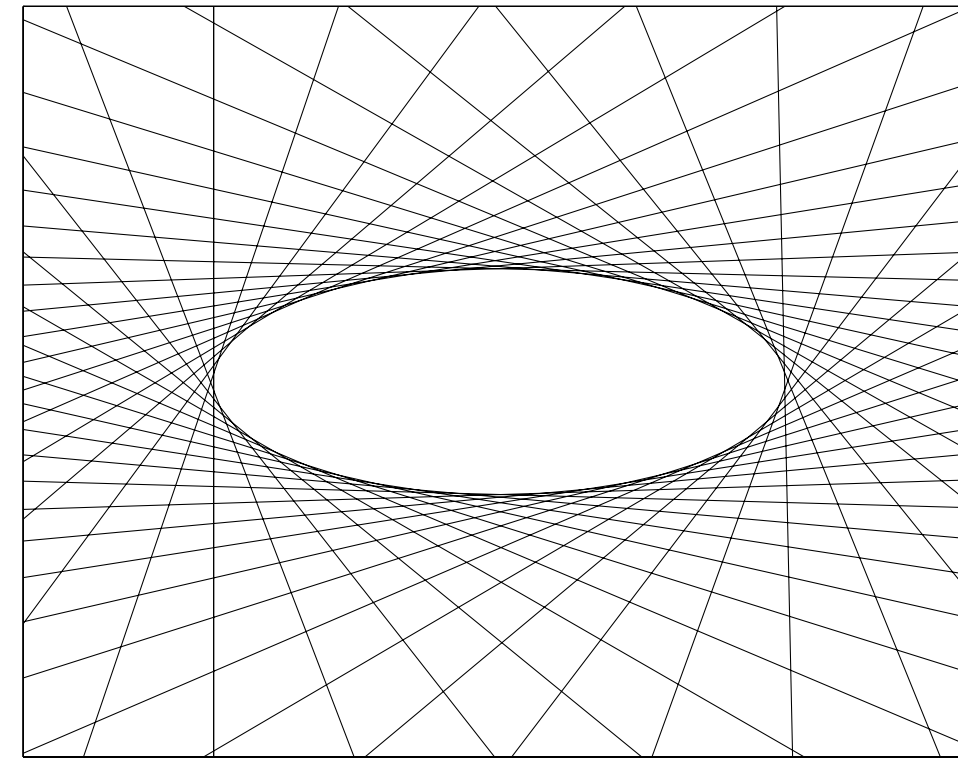
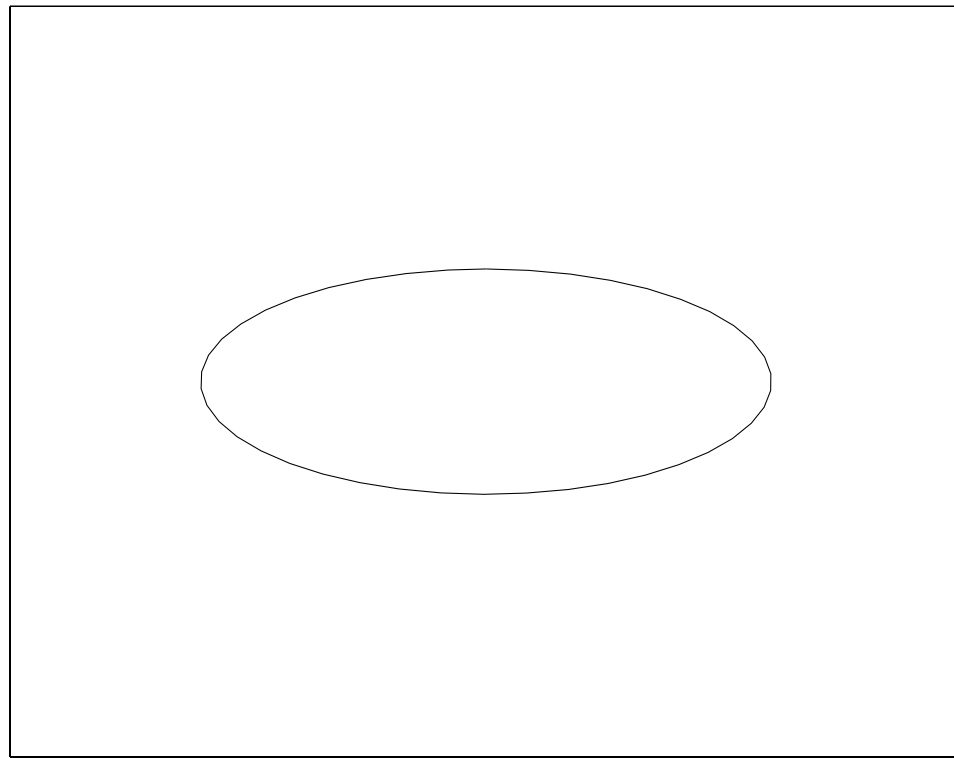


Figure courtesy: Hartley & Zisserman

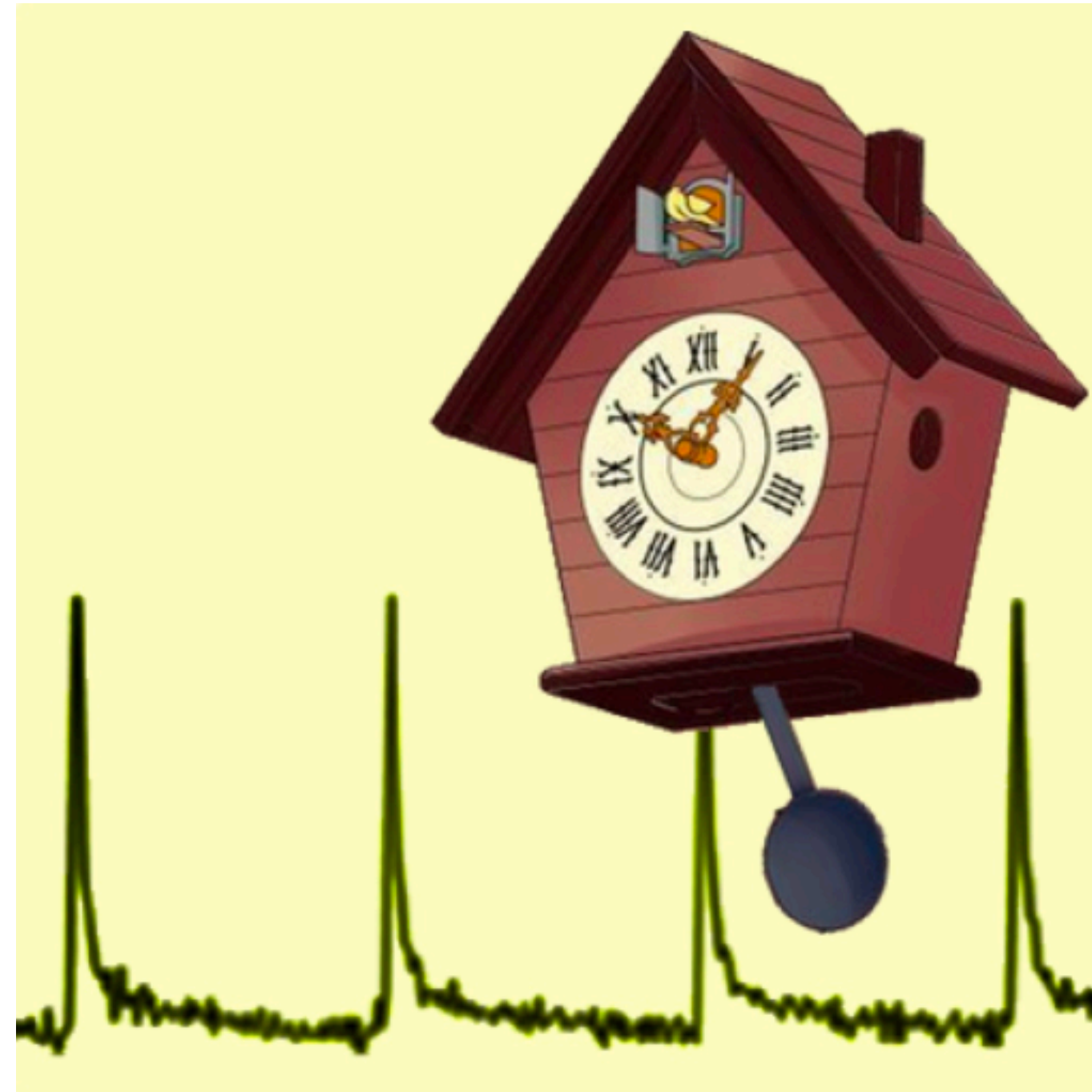
A reminder to ask questions!

CAN UNICORNS ROAST
MARSHMALLOWS?

No.

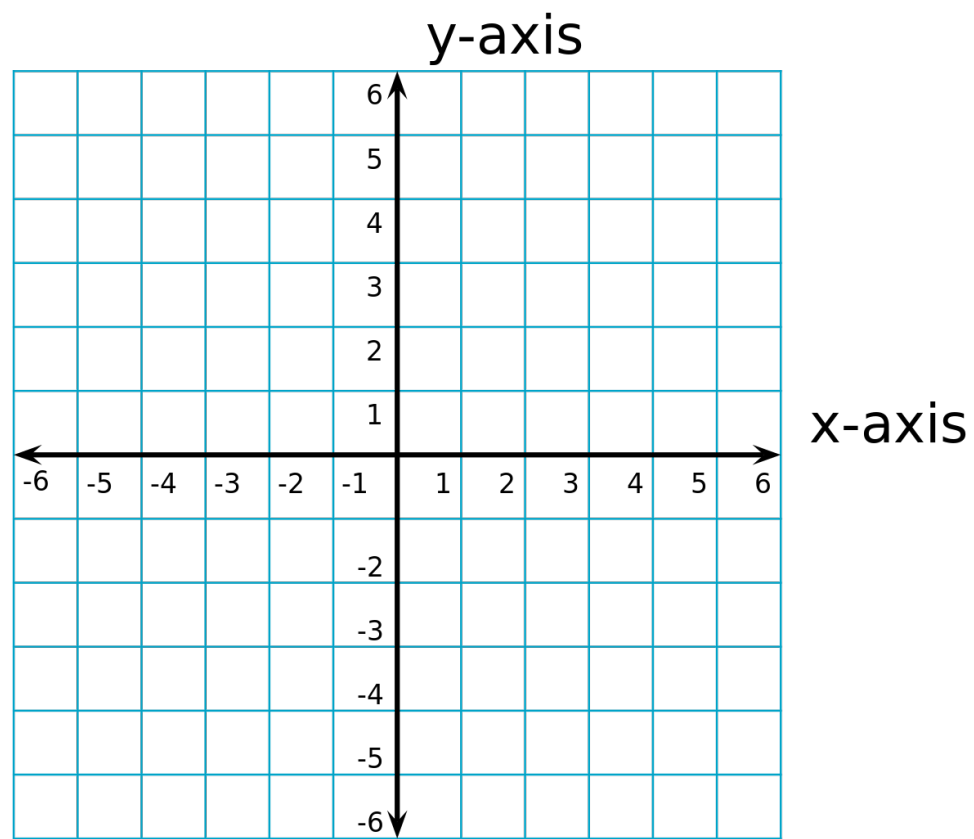


Disclaimer: Calibration ongoing!



Let me know if we are going too fast (or too slow!)

Coordinate Spaces $\mathbb{R}^n$

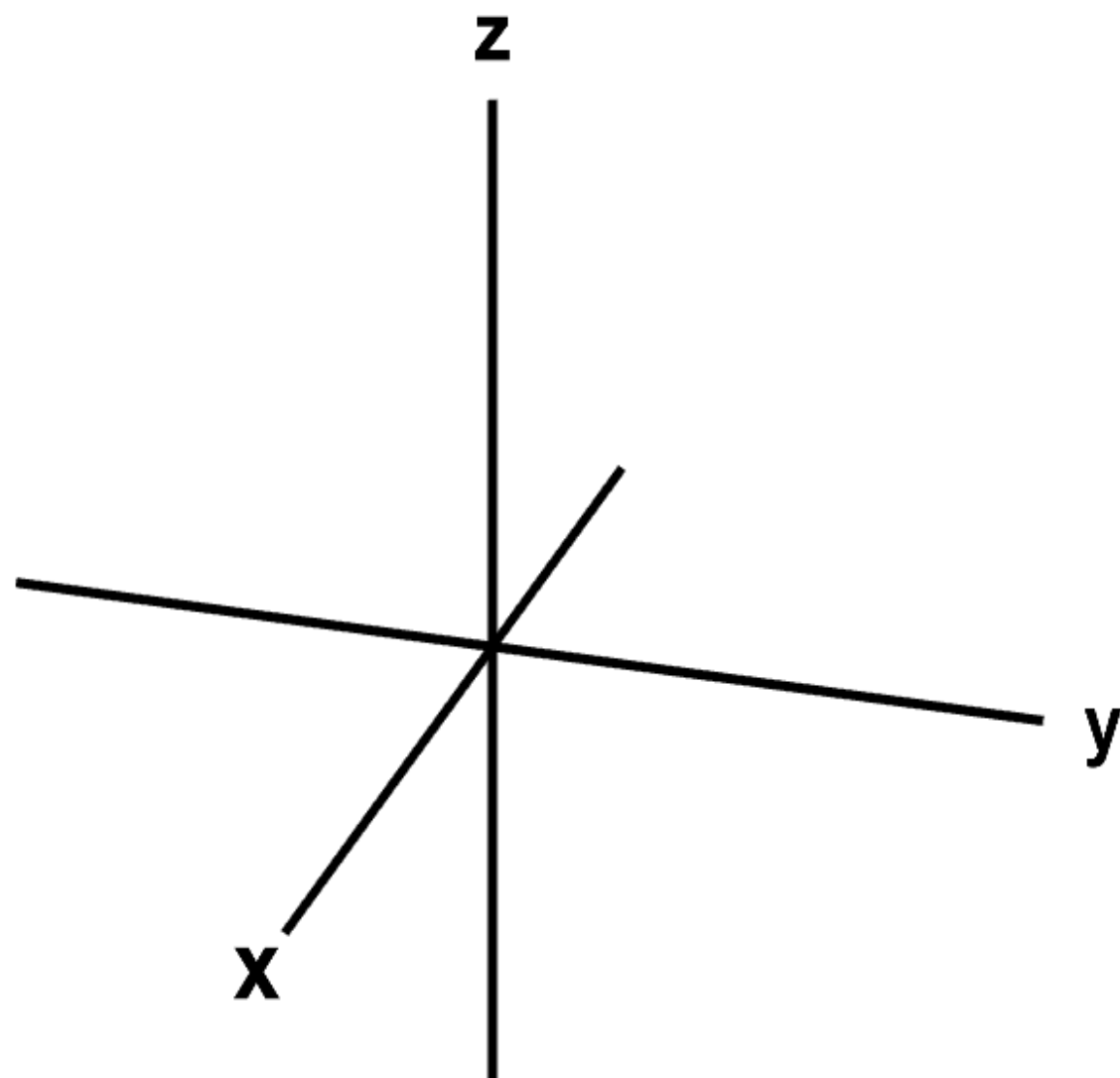


$$\tilde{\mathbf{x}} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Used to represent points and vectors

Points at infinity not represented

(but why is this a big deal?)



$$\tilde{\mathbf{x}} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

```
def line_intersection(line1, line2):
    xdiff = (line1[0][0] - line1[1][0], line2[0][0] - line2[1][0])
    ydiff = (line1[0][1] - line1[1][1], line2[0][1] - line2[1][1])

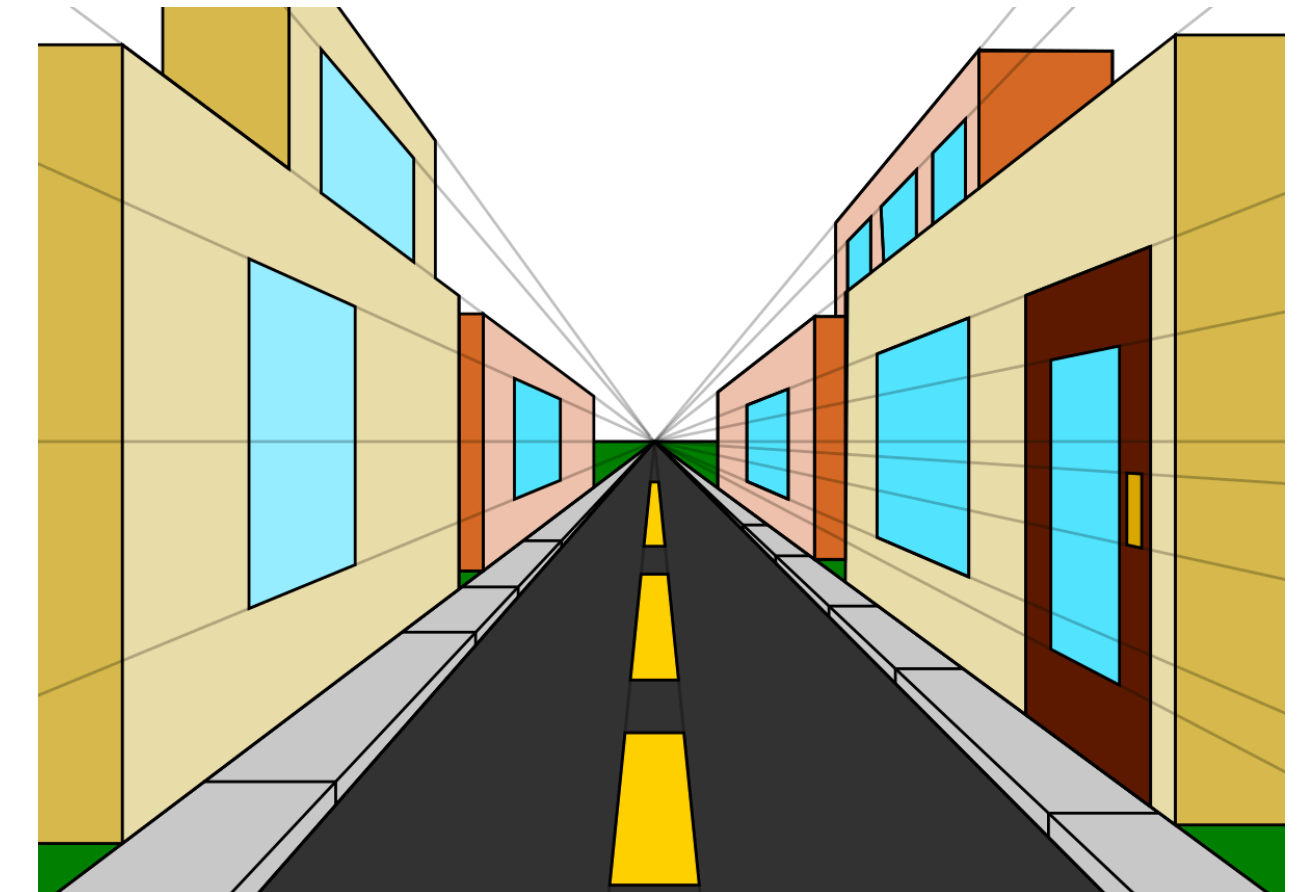
    def det(a, b):
        return a[0] * b[1] - a[1] * b[0]

    div = det(xdiff, ydiff)
    if div == 0:
        raise Exception('lines do not intersect')

    d = (det(*line1), det(*line2))
    x = det(d, xdiff) / div
    y = det(d, ydiff) / div
    return x, y

print line_intersection((A, B), (C, D))
```

special cases frequently arise in
common (finite) operations



talking about points at infinity
is sometimes important

Homogenous Coordinates $\mathbb{P}^n$

$\mathbb{R}^2$ 'Standard' 2D space represented by points (x, y)

$\mathbb{P}^2$ Projective 2D space represented by (x, y, z) s.t. $(x, y, z) \equiv (\lambda x, \lambda y, \lambda z)$

e.g. describing a line passing through origin via a point on the line (a 3D representation for a 2 DoF entity)

Q: Starting at a 2D point (x, y) in euclidean space, where do we reach if we keep walking in direction (u, v) ?

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} u \\ v \\ 0 \end{bmatrix} \equiv \begin{bmatrix} \frac{1}{\lambda}x + u \\ \frac{1}{\lambda}y + v \\ \frac{1}{\lambda} \end{bmatrix} \rightarrow \begin{bmatrix} u \\ v \\ 0 \end{bmatrix}$$

Q: What point do we reach if walking in 'opposite' direction?

Points in 2D Projective Space

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix} \equiv \begin{bmatrix} x/w \\ y/w \\ 1 \end{bmatrix}$$

A vector with 3 numbers is just a representation of a point (among other possible representations of the same point)

Q (True/False):

$w = 0$ for points at infinity

$$\begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$$

is a unique representation for the corresponding point at infinity

Lines in 2D Projective Space

$$\mathbb{R}^2$$

$$(\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2)$$

Points on a line:

$$\alpha \tilde{\mathbf{x}}_1 + (1 - \alpha) \tilde{\mathbf{x}}_2$$

Line Equation:

$$ax + by + c = 0$$

$$\mathbb{P}^2$$

$$(\mathbf{x}_1, \mathbf{x}_2)$$

$$\alpha \mathbf{x}_1 + \beta \mathbf{x}_2$$

$$ax + by + cw = 0$$

$$\mathbf{l}^\top \mathbf{x} = 0$$

Lines in 2D Projective Space

$$\mathbf{l}^T \mathbf{x} = 0$$

Q: Find line $\mathbf{l}$ intersecting points $\mathbf{x}_1$ and $\mathbf{x}_2$.

$$\mathbf{l}^T \mathbf{x}_1 = 0 \quad \mathbf{l}^T \mathbf{x}_2 = 0$$

$$\mathbf{l} = \mathbf{x}_1 \times \mathbf{x}_2$$

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= [\mathbf{a}]_{\times} \mathbf{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \\ \mathbf{a} \times \mathbf{b} &= [\mathbf{b}]_{\times}^T \mathbf{a} = \begin{bmatrix} 0 & b_3 & -b_2 \\ -b_3 & 0 & b_1 \\ b_2 & -b_1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, \end{aligned}$$

Cool trick (really useful later): cross product as matrix multiplication!

Lines in 2D Projective Space: Duality

$$\mathbf{l}^\top \mathbf{x} = 0$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

$$\mathbf{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Both invariant to scalar multiplication

Informally: For any theorem/construct about points, there will be a corresponding one about lines

Result 2.6. Duality principle. *To any theorem of 2-dimensional projective geometry there corresponds a dual theorem, which may be derived by interchanging the roles of points and lines in the original theorem.*

Lines in 2D Projective Space

$$\mathbf{l}^T \mathbf{x} = 0$$

Q: Find line $\mathbf{l}$ intersecting points $\mathbf{x}_1$ and $\mathbf{x}_2$.

$$\mathbf{l} = \mathbf{x}_1 \times \mathbf{x}_2$$

Q: Find intersection point $\mathbf{x}$ of lines $\mathbf{l}_1$ and $\mathbf{l}_2$.

$$\mathbf{x} = \mathbf{l}_1 \times \mathbf{l}_2$$

Exercise: show that parallel lines intersect at infinity

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

Lines in 2D Projective Space

$$\mathbf{l}^\top \mathbf{x} = 0$$

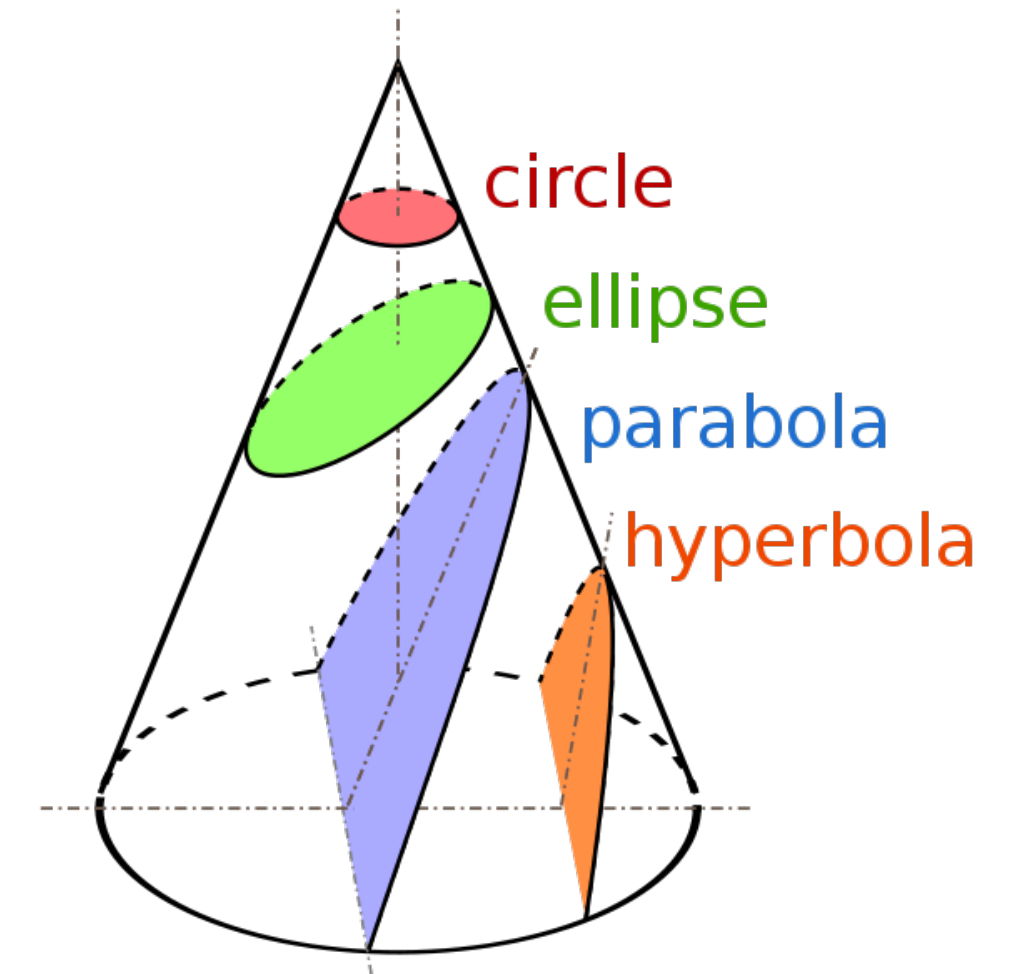
All points at infinity lie on a line

$$\mathbf{l}_\infty = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Conics in 2D Projective Space

$\mathbb{R}^2$

$$ax^2 + by^2 + cxy + dx + ey + f = 0$$



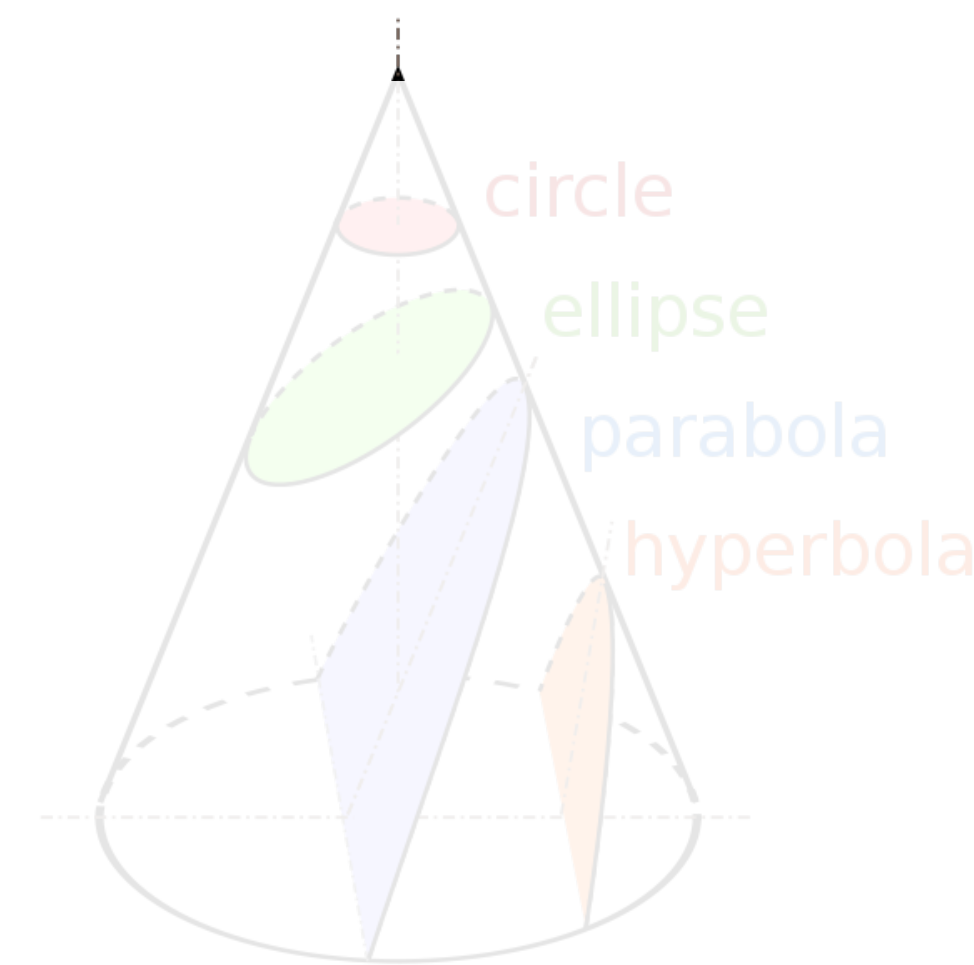
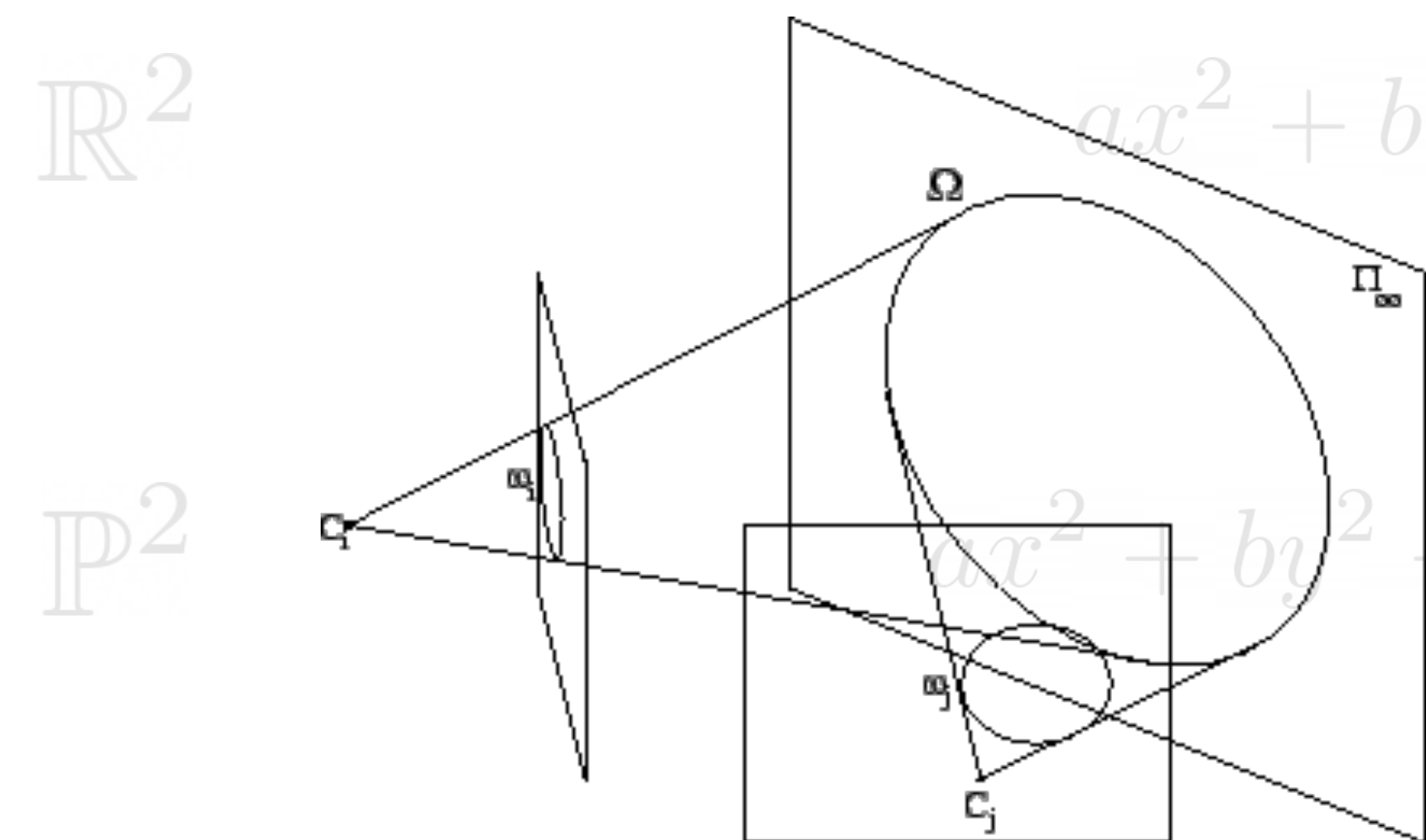
$\mathbb{P}^2$

$$ax^2 + by^2 + cxy + dxw + eyw + fw^2 = 0$$

$$\mathbf{x}^T C \mathbf{x} = 0$$

$$C = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}$$

Conics in 2D Projective Space



Deeply related to camera calibration
(later lectures)

$$\mathbf{x}^T C \mathbf{x} = 0$$

$$C = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}$$

Why do we care about conics?

Conics in 2D Projective Space

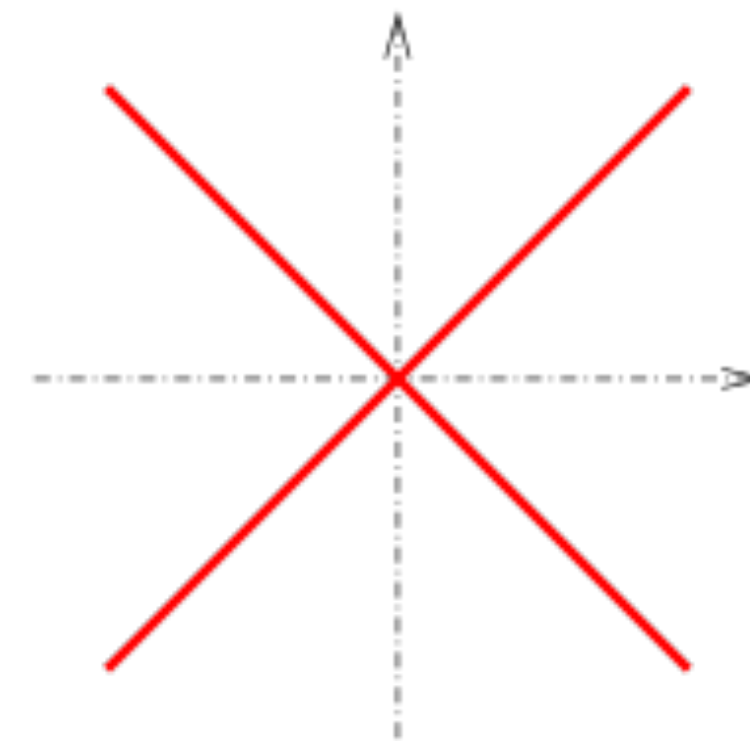
$$\mathbf{x}^T C \mathbf{x} = 0$$

Each symmetric matrix C defines a conic!

degenerate iff $\det(C) = 0$

Example: conic consisting of points on 2 lines $\mathbf{l}, \mathbf{m}$

$$C = \mathbf{l}\mathbf{m}^T + \mathbf{m}\mathbf{l}^T$$



Exercise: show that points not on $\mathbf{m}$ or $\mathbf{l}$ do not lie on C

Conics in 2D Projective Space

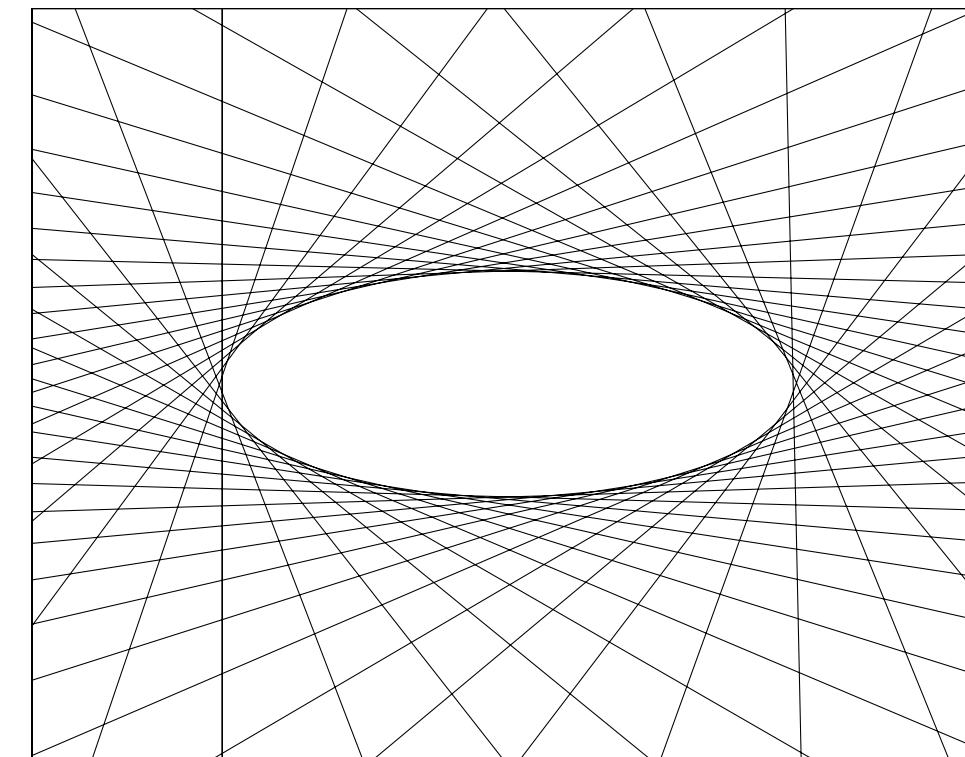
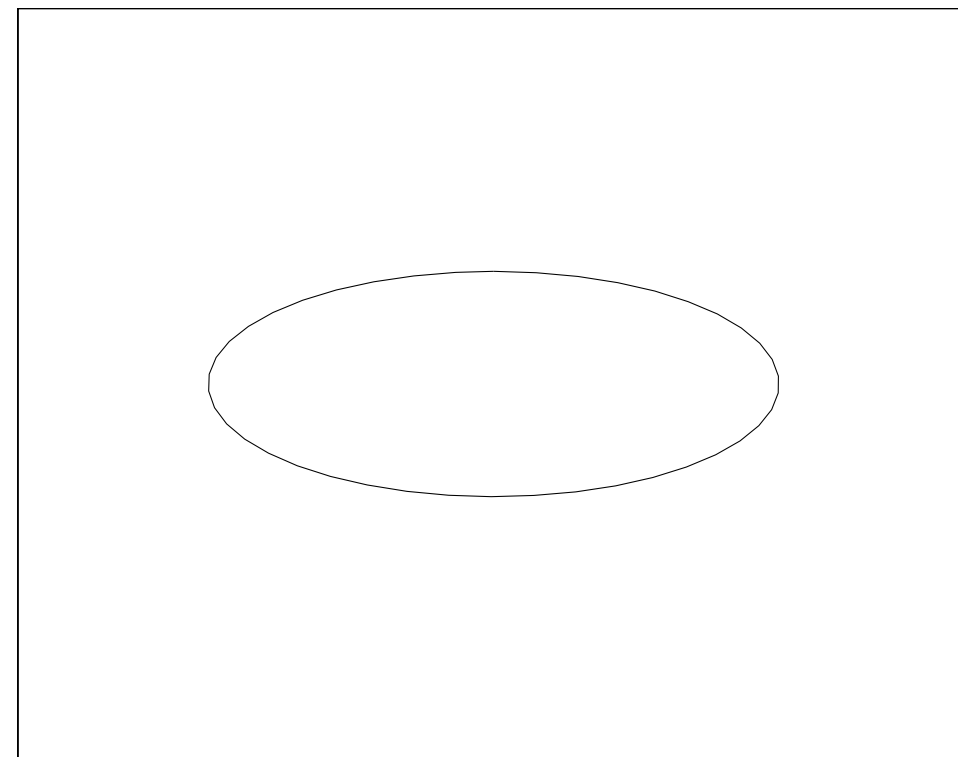
$$\mathbf{x}^T C \mathbf{x} = 0$$

Given a point $\mathbf{x}$ on conic C , $C\mathbf{x}$ is the tangent to C at $\mathbf{x}$

(easy to see that $\mathbf{x}$ lies on this line, but slightly tricky to prove no other $\mathbf{y}$ on lines on C and $C\mathbf{x}$)

Dual Conic C^* :
defines the set of
lines tangent to C

$$\mathbf{l}^T C^* \mathbf{l} = 0$$



$$C^* = C^{-1} \text{ (up to scale)}$$

(adjoint of C if $\det(C) = 0$)

Conics in 2D Projective Space

$$\mathbf{x}^T C \mathbf{x} = 0$$

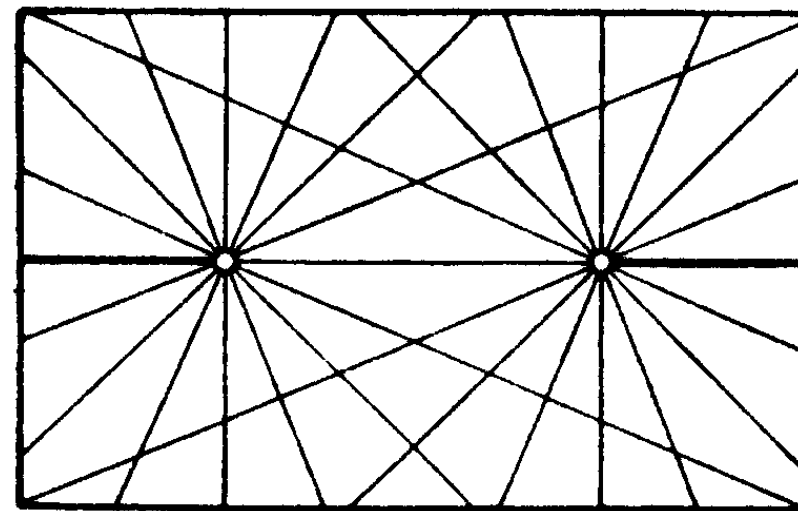
point conic

$$\mathbf{l}^T C^* \mathbf{l} = 0$$

line conic

Example of degenerate line conic: all lines passing through
points $\mathbf{x}, \mathbf{y}$

$$C^* = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$$



may seem abstract, but we'll see a very useful special case later — for calibration, rectification etc.!

Questions?